Worksheet 1-14 A Infinite Limits and Vertical Asymptotes #1
AP Calculus Name_____

Directions: Evaluate the following limits or explain why the limit does not exist.

1. $\lim_{x\to 4^+} \frac{1}{(x-4)^2}$

2. $\lim_{x\to 4^-} \frac{1}{(x-4)^2}$

3. $\lim_{x\to 2^-} \frac{x+3}{x-2}$

4. $\lim_{x\to 0^+} \frac{x^2+1}{2x}$

5. $\lim_{x\to -2} \frac{4-3x}{x+2}$

6. $\lim_{x\to 0} \frac{3x+2}{x^2}$

7. $\lim_{x\to 2} \frac{x^2-7x+10}{x^2+2x-8}$

8. $\lim_{x\to 4^-} \frac{x-4}{3|x-4|}$

9. $\lim_{x\to -5} \frac{x^2+4x-5}{x^2+10x+25}$

10. Write a function that has a vertical asymptote at $x = 3$ and at $x = -1$.

11. Write a function that has a vertical asymptote at $x = -2$ and a hole at $x = 7$.

12. The function $h(x)$ decreases without bound as x approaches 8. Write a limit statement to model this behavior.